

ORACLE

Cloud Operations for Exadata Database Service on Exascale Infrastructure

Change Operations Scenarios

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Office of CTO | EMEA Technology Engineering
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Agenda

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Introduction

Introduction



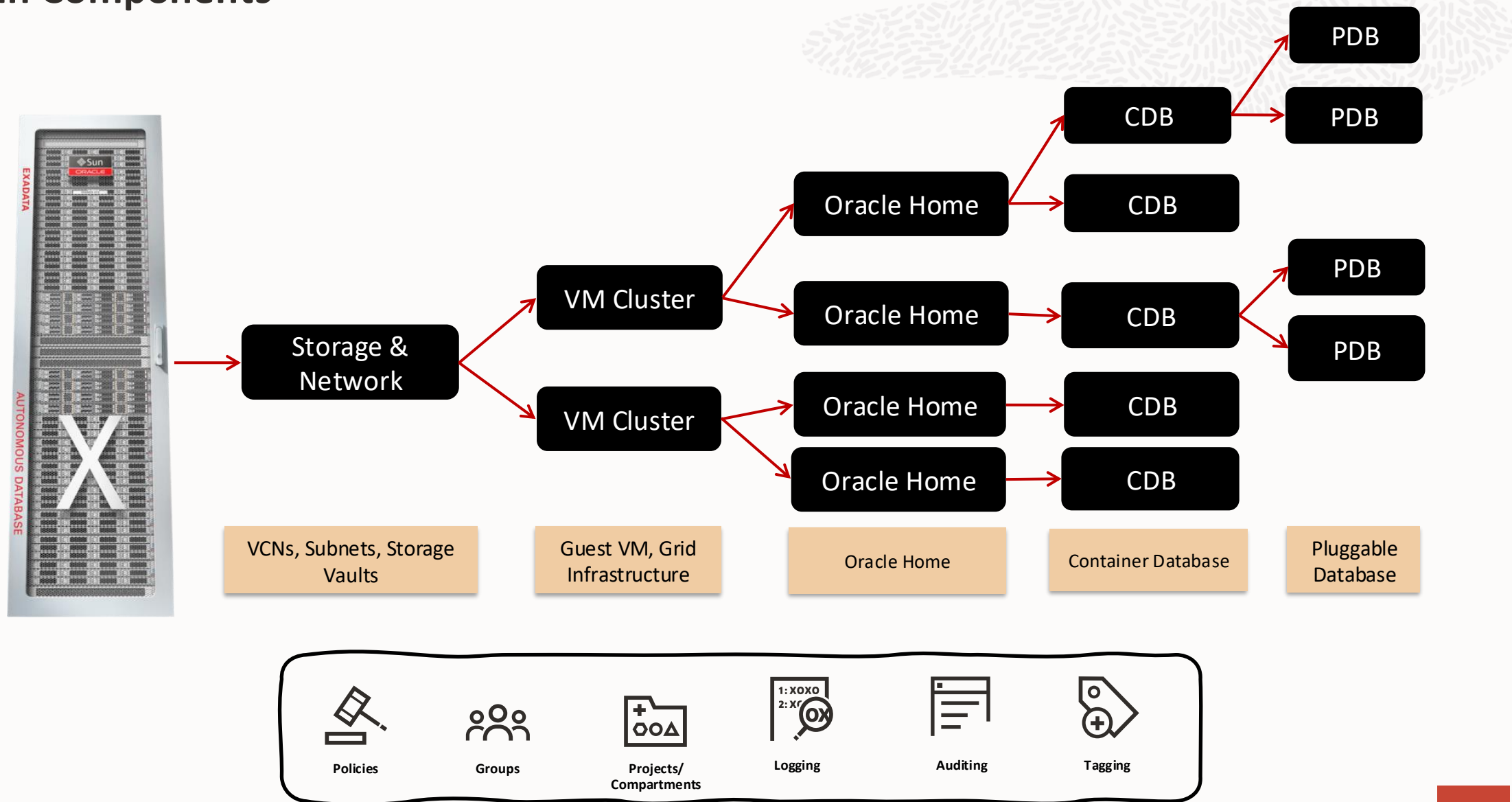
This Workshop provides information about Exadata Database Service on Exascale Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VC, OH, DBs, DG, scaling and patching

The code in this Workshop is just sample code and can be incomplete.

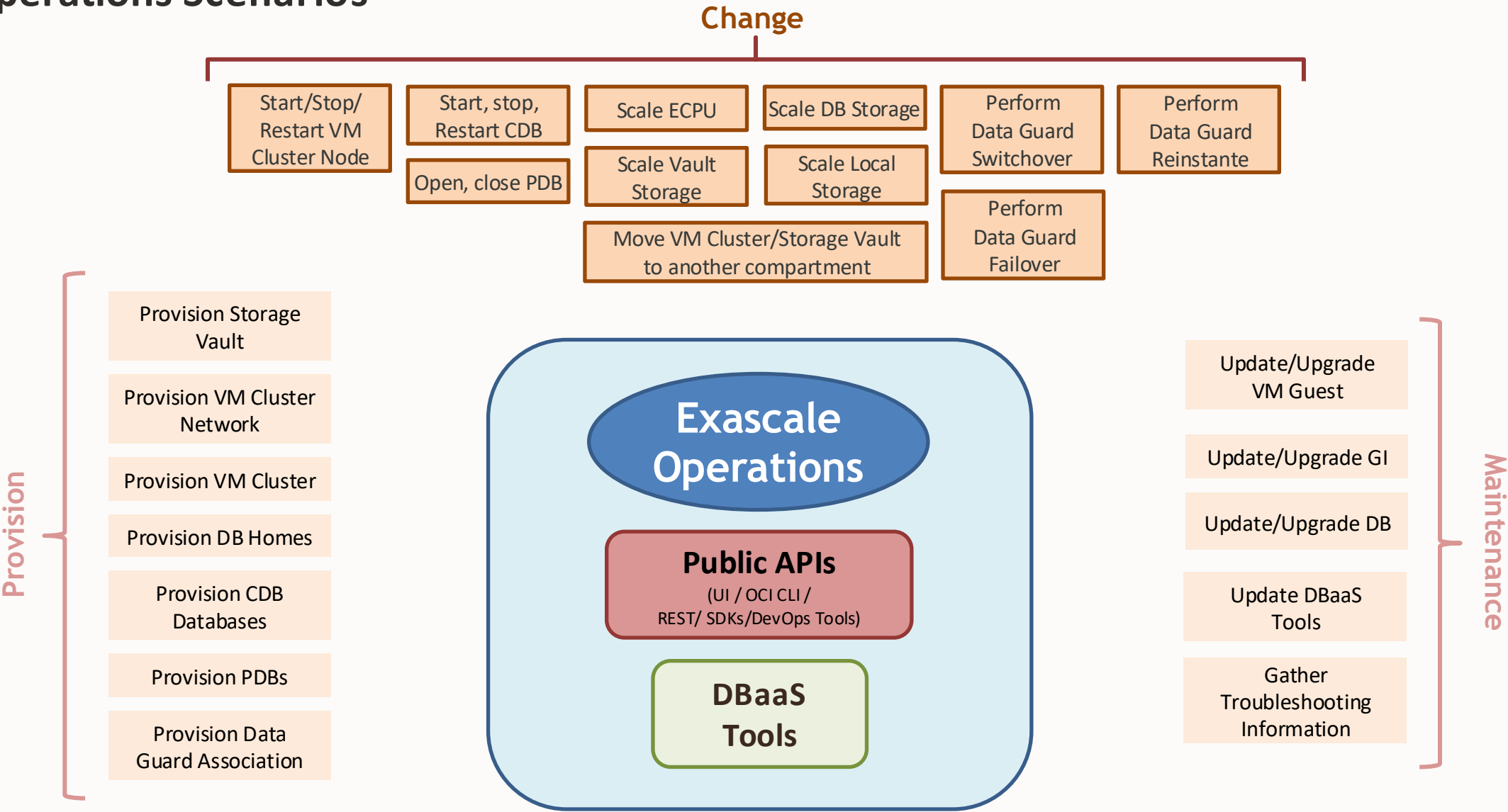
To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

Main Components



Change Operations

Operations Scenarios



Change compartment

Storage Vault - OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure →
Exascale Storage Vault → Select Actions (3 dots) on the Storage Vault → **Move resource**

Command Line Interface (OCI CLI)

```
oci db exascale-db-storage-vault change-compartment --compartment-id <compId> --exascale-db-storage-vault-id <Storage-Vault-Id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

change_exascale_db_storage_vault_compartment_response =
database_client.change_exascale_db_storage_vault_compartment(

change_exascale_db_storage_vault_compartment_details=oci.database.model
s.ChangeExascaleDbStorageVaultCompartmentDetails(
    compartment_id="ocidl.test.oc1..<compartmentId>",
    exascale_db_storage_vault_id="ocidl.test.oc1..<DbStorageVaultId>",
    opc_retry_token="EXAMPLE-opcRetryToken-Value",
    opc_request_id="I2FEWWXJWXV5HBZCE6JF<unique_ID>",
    if_match="EXAMPLE-ifMatch-Value")
print(change_exascale_db_storage_vault_compartment_response.headers)
```

OCI APIs

```
POST
/20160918/exascaleDbStorageVaults/{VaultId}/actions/changeCompartment
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

VM Cluster - OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select Actions (3 dots) on the VM Cluster → **Move resource**

Command Line Interface (OCI CLI)

```
oci db exadb-vm-cluster change-compartment --compartment-id <compId> --exadb-vm-cluster-id <vmClId> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

change_exadb_vm_cluster_compartment_response =
database_client.change_exadb_vm_cluster_compartment(
change_exadb_vm_cluster_compartment_details=oci.database.models.ChangeE
xadbVmClusterCompartmentDetails(
    compartment_id="ocidl.test.oc1..<compId>",
    exadb_vm_cluster_id="ocidl.test.oc1..<exadbVmClusterId>",
    opc_retry_token="EXAMPLE-opcRetryToken-Value",
    opc_request_id="CKE5MSY3FUIFK9KDZBWB<unique_ID>",
    if_match="EXAMPLE-ifMatch-Value")
print(change_exadb_vm_cluster_compartment_response.headers)
```

OCI APIs

```
POST /20160918/exadbVmClusters/{exadbVmClId}/actions/changeCompartment
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Scale Storage Vault

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → Exascale Storage Vault → Select Storage Vault → **Scale storage vault**

Command Line Interface (OCI CLI)

```
oci db exascale-db-storage-vault update --exascale-db-storage-vault-id <exascale-db-storage-vault-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

update_exascale_db_storage_vault_response =
database_client.update_exascale_db_storage_vault(
    exascale_db_storage_vault_id="ocidl.test.oc1..<exasDbStVaultId>",
    update_exascale_db_storage_vault_details =
oci.database.models.UpdateExascaleDbStorageVaultDetails(
    display_name="EXAMPLE-displayName-Value",
    description="EXAMPLE-description-Value",
    high_capacity_database_storage =
oci.database.models.ExascaleDbStorageInputDetails(
    total_size_in_gbs=477),
    ...),
    ...)
print(update_exascale_db_storage_vault_response.data)
```

OCI APIs

```
PUT /20160918/exascaleDbStorageVaults/<storageVaultId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "highCapacityDatabaseStorage": {
    "capacitySizeInGbs": 256
  },
  "additionalFlashCache": 20,
  displayName: "exascaleVaultName"
}
```

VM Cluster Scale

What can be scaled?

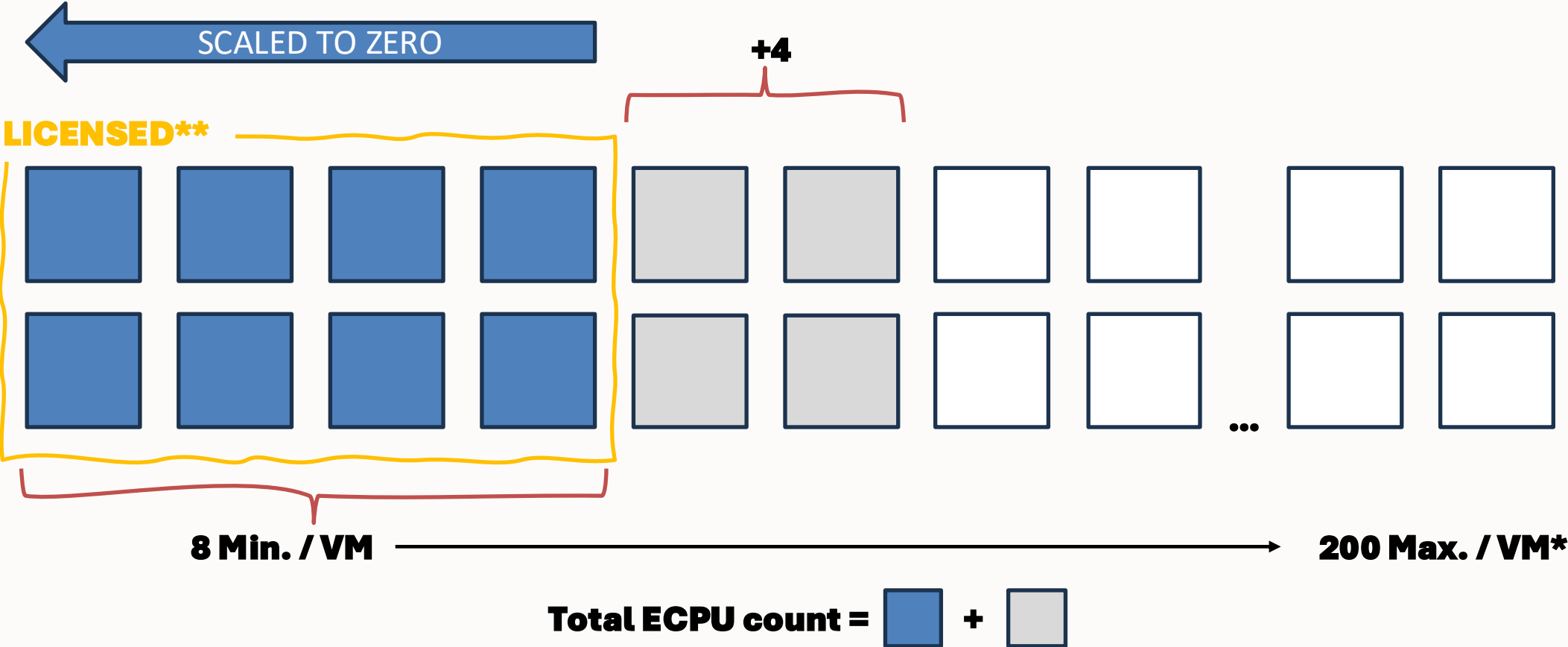


- **ECPU**
- **Local File Storage**
- **DB Storage**
- **Memory?** Memory is calculated based on 11 GB per total cores.
- You can scale VM Clusters horizontally by adding or removing VMs to or from an existing VM Cluster.
 - Add a VM to a VM Cluster
 - This operation is only available with Multi-VM enabled Infrastructure.
 - Requires that all TCP ports are open for the client subnet CIDR for ingress and egress.
 - The VM that is added will have the same resources as the other VMs in the cluster.
 - Add a VM to a VM Cluster is NOT supported using Terraform.

Core Reservation and Scaling

- Enabled ECPU
- Reserved ECPU
- Non reserved ECPU

An ECPU is an abstracted measure of compute resources based on the number of cores elastically allocated from a pool of compute servers.



15 **Billed by the second, with a minimum usage period of 1 minute

* Or limited by the number of total ECPUs specified for the VM



OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Select VM Cluster → **Scale VM Cluster (ECPUs/Memory/VM File system storage)**

Command Line Interface (OCI CLI)

```
oci db exadb-vm-cluster update --exascale-vm-cluster-id <exascale-vm-cluster-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

update_exadb_vm_cluster_response =
database_client.update_exadb_vm_cluster(
    exadb_vm_cluster_id="ocidl.test.oc1..<exadbVmClusterId>",
    update_exadb_vm_cluster_details =
oci.database.models.UpdateExadbVmClusterDetails(
    display_name="EXAMPLE-displayName-Value",
    total_e_cpu_count=971,
    enabled_e_cpu_count=380,
    vm_file_system_storage =
oci.database.models.ExadbVmClusterStorageDetails(
    total_size_in_gbs=139),
    node_count=118,
    ...),
    ...)
print(update_exadb_vm_cluster_response.data)
```

OCI APIs

```
PUT /20160918/exadbVmClusters/<exadbVmClusterId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "enabledECpuCount" : 10,
  "licenseModel" : "LICENSE_INCLUDED",
  "dbCertificateId" : "ocidl.dbcertificate.oc1."
}
```


Start/Stop VM Cluster Node

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Virtual Machines → Select Actions (3 dots) on the VM → **Start/Stop**

Command Line Interface (OCI CLI)

```
oci db node start/stop/reset/soft-reset --db-node-id <node-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

db_node_action_response = database_client.db_node_action(
    db_node_id="ocid1.test.oc1..<unique_ID>EXAMPLE-dbNodeId-Value",
    action="RESET",
    opc_retry_token="EXAMPLE-opcRetryToken-Value",
    if_match="EXAMPLE-ifMatch-Value")
print(db_node_action_response.data)
```

OCI APIs

```
POST /20160918/dbNodes/<dbNodeId>?action=<action>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Performs one of the following power actions on the specified DB node:

- **start** - power on
- **stop** - power off gracefully
- **softreset** - ACPI shutdown and power on
- **reset** - power off and power on

Start/Stop CDB

Local VM Command-Line Interfaces - dbaascli



Start a specific database

```
$ dbaascli database start [--dbname <value>] [--mode <value>]
```

Stop a specific database

```
$ dbaascli database stop [--dbname <value>] [--mode <value>]
```

Bounce a specific database

```
$ dbaascli database bounce [--dbname <value>] [--rolling <value>]
```

Checks status of a database

```
$ dbaascli database status [--dbname <value>] [--service <value> --user <value>]  
[INFO] [DBAAS-14011] - The usage of this command is deprecated.  
ACTION: It is recommended to use 'dbaascli database getDetails' for this operation.
```

¹... Not all parameters listed. Please check the command definition

Open/Close PDB

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Container databases → Select CDB → Pluggable Databases → Select Actions (3 dots) on the PDB → **Start/Stop**

Command Line Interface (OCI CLI)

Start PDB

```
oci db pluggable-database start --pluggable-database-id <pluggable-database-id> [OPTIONS]
```

Stop PDB

```
oci db pluggable-database stop --pluggable-database-id <pluggable-database-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

start_pluggable_database_response =
database_client.start_pluggable_database(
    pluggable_database_id="ocid1.test.oc1..<PDBId>", ...)
print(start_pluggable_database_response.data)

stop_pluggable_database_response =
database_client.stop_pluggable_database(
    pluggable_database_id="ocid1.test.oc1..<PDBId>", ...)
print(stop_pluggable_database_response.data)
```

OCI APIs

```
POST /20160918/pluggableDatabases/<pluggableDatabaseId>/actions/stop
POST /20160918/pluggableDatabases/<pluggableDatabaseId>/actions/start
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Local VM Command-Line Interfaces - dbaascli

Open a PDB

```
$ dbaascli pdb open {--pdbName <value> | --pdbUID <value>} --dbname <value> [--openMode <value>]
[--startServices <value>] [--waitForCompletion <value>] [--setPDBRefreshModeNone [--skipPDBRefresh]
[--pdbAdminUserName <value>]] [--executePrereqs] [--resume [--sessionID <value>]] [--blobLocation <value>]
```

Close a PDB

```
$ dbaascli pdb close {--pdbName <value> | --pdbUID <value>} --dbname <value> [--waitForCompletion <value>]
```

Bounce a PDB

```
$ dbaascli pdb bounce {--pdbName <value> | --pdbUID <value>} --dbname <value> [--openMode <value>]
[--startServices <value>] [--waitForCompletion <value>]
```

To get info...

Display details about a specific PDB

```
$ dbaascli pdb getDetails --dbname <value> {--pdbName <value> | --pdbUID <value>}
```

Get connectionString of the specified PDB

```
$ dbaascli pdb getConnectionString --dbname <value> {--pdbName <value> | --pdbUID <value>}
```

¹... Not all parameters listed. Please check the command definition

DataGuard Changes

Switchover

A **switchover** is a role reversal between the primary database and one of its standby databases.

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Container databases → Select CDB → Data Guard Associations → Select Actions (3 dots) on the Member → **Switchover**

Command Line Interface (OCI CLI)

```
oci db data-guard-association switchover --data-guard-association-id <data-guard-association-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

switchover_data_guard_association_response =
database_client.switchover_data_guard_association(
    database_id="ocidl.test.oc1..<databaseId>",
    data_guard_association_id="ocidl.test.oc1..<DGAssId>",
    switchover_data_guard_association_details =
oci.database.models.SwitchoverDataGuardAssociationDetails(
    database_admin_password="DBAdmPwd"),
    if_match="EXAMPLE-ifMatch-Value")
print(switchover_data_guard_association_response.data)
```

OCI APIs

```
POST /20160918/databases/<databaseId>/actions/switchover
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

dbaascli

```
$ dbaascli dataguard switchover --dbname <value> [--targetStandbyDBUniqueName <value>] [--executePrereqs] [--
enabledDGDebug] [--waitForCompletion <value>] [--resume [--sessionID <value>]]
```

Failover

A **failover** is a role transition in which one of the standby databases is transitioned to the primary role after a primary outage

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Container databases → Select CDB → Data Guard Associations → Select Actions (3 dots) on the Member → **Failover**

Command Line Interface (OCI CLI)

```
oci db data-guard-association failover --data-guard-association-id <data-guard-association-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

failover_data_guard_association_response =
database_client.failover_data_guard_association(
    database_id="ocidl.test.oc1.<databaseId>",
    data_guard_association_id="ocidl.test.oc1.<DGAssId>",
    failover_data_guard_association_details =
oci.database.models.FailoverDataGuardAssociationDetails(
    database_admin_password="DBAdmPwd"),
    if_match="EXAMPLE-ifMatch-Value")
print(failover_data_guard_association_response.data)
```

OCI APIs

```
POST /20160918/databases/<databaseId>/actions/failover
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

dbaascli

```
$ dbaascli dataguard failover --dbname <value> [--useImmediateFailover] [--executePrereqs] [--waitForCompletion  
<value>] [--resume [--sessionID <value>]]
```

Reinstate

To **reinstate** is to restore a failed or disabled primary database back to a valid standby role after a failover.

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters → Container databases → Select CDB → Data Guard Associations → Select Actions (3 dots) on the Member → **Reinstate**

Command Line Interface (OCI CLI)

```
oci db data-guard-association reinstate --data-guard-association-id <data-guard-association-id> [OPTIONS]
```

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)

reinstate_data_guard_association_response =
database_client.reinstate_data_guard_association(
    database_id="ocidl.test.oc1..<databaseId>",
    data_guard_association_id="ocidl.test.oc1..<DGAssId>",
    reinstate_data_guard_association_details =
oci.database.models.ReinstateDataGuardAssociationDetails(
    database_admin_password="DBAdmPwd"),
    if_match="EXAMPLE-ifMatch-Value")
print(reinstate_data_guard_association_response.data)
```

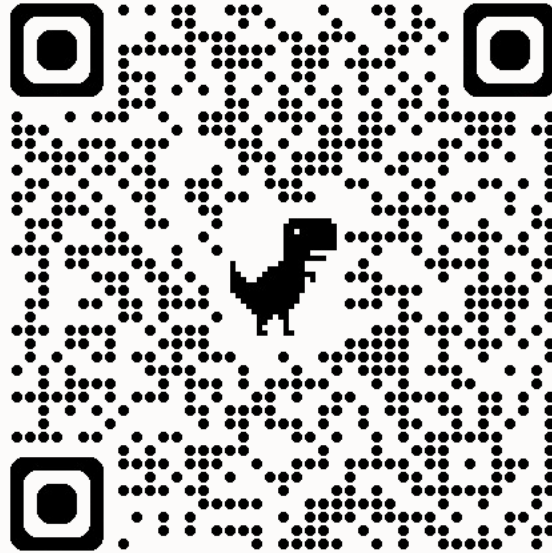
OCI APIs

```
POST /20160918/databases/<databaseId>/actions/reinstate
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

dbaascli

```
$ dbaascli dataguard reinstate --dbname <value> [--primaryDBUniqueName <value>] [--executePrereqs] [--
waitForCompletion <value>] [--resume [--sessionID <value>]]
```

Stay tuned



Operations Advisory
GitHub Repository



Thank you

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